SCD-MoE: Adaptive Expert Routing for Efficient Remote Sensing Semantic Change Detection

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***Abstract*—Semantic change detection (SCD) aims to simultaneously localize changed regions and identify land-cover transitions from bi-temporal images. However, existing methods usually process bi-temporal features with fixed interaction structures or shared fusion functions, providing limited flexibility to explicitly adapt interaction preferences to diverse change patterns. In addition, SCD requires both change localization and semantic recognition, which emphasize different information cues: the former relies more on discrepancy-sensitive features, while the latter requires semantic-preserving and region-consistent representations, yet such task-specific preferences are often insufficiently modeled. To address these challenges, we propose SCD-mixture-of-expert (MoE), an efficient dynamic framework that leverages conditional computation to model sample-adaptive interactions and task-specific feature preferences. First, we introduce an interaction-adaptive MoE (IA-MoE), which routes bi-temporal features to difference-aware and semantic-preserving experts for adaptive interaction modeling. Through selective expert weighting, IA-MoE enhances change-related cues while maintaining semantic context. Second, we develop a task-aware routing MoE (TAR-MoE), which employs task-specific gates over a shared expert pool to generate localization- and recognition-oriented representations. This design enables controlled information sharing while allowing each task to emphasize suitable feature transformations. Experiments on three benchmarks show that SCD-MoE consistently achieves superior performance, including a 5.04% separated Kappa coefficient (SeK) improvement on Landsat-SCD, while using only 48% of the parameters and substantially fewer FLOPs compared with the previous state-of-the-art method. Code and pretrained models will be made available at https://github.com/HuQ1an/SCD-MoE**

***Index Terms*—Dynamic routing, mixture of experts (MoE), remote sensing, semantic change detection (SCD).**

I. Introduction

MONITORING dynamic changes on the Earth's surface is essential for understanding our planet's evolution [1], [2], [3], [4]. As a key technique in this field, semantic change detection (SCD) [5], [6], [7] aims to provide a detailed interpretation of surface dynamics by simultaneously localizing changed regions and identifying their semantic changes. Depending on available observations and task requirements, SCD may exploit multitemporal or multimodal data, as well as auxiliary geospatial information, to describe different forms of semantic change [8], [9], [10], [11]. These changes may involve land-cover transitions, object-state variations, or disaster-related damage levels. Owing to its comprehensive analytic capability, SCD has become indispensable in diverse applications, ranging from urban expansion monitoring [12], [13] and disaster damage assessment [14], [15] to resource management [16], [17], [18].

Given the broad availability of optical observations, SCD based on bi-temporal optical imagery has become one of the most widely studied formulations. Reliable change interpretation therefore requires effective mining and comparison of information across the two temporal phases. Driven by this necessity, researchers have sought to enhance SCD performance by designing sophisticated bi-temporal interaction mechanisms to capture spatiotemporal dependencies. Existing methods have explored various interaction strategies, such as feature differencing, concatenation, attention-based fusion, and gated fusion [19], [20], [21], [22], [23], [24]. Although many existing bi-temporal interaction strategies can generate input-dependent responses, their overall interaction formulation is usually predefined and shared across samples. In other words, different samples are generally processed through the same type of interaction pathway. This may limit their flexibility in accommodating diverse change patterns characterized by varying appearance conditions, semantic complexity, and occurrence frequencies. Therefore, a sample-adaptive bi-temporal interaction paradigm is desirable, enabling the model to adjust its interaction behavior according to input-specific change characteristics.

In addition, SCD jointly involves change localization and semantic recognition, which are closely related but emphasize different feature properties. The binary change detection branch relies more on discrepancy-sensitive and boundary-aware cues to localize changed regions, whereas semantic recognition requires class-discriminative and region-consistent representations to preserve land-cover semantics. Existing decoders usually follow either multibranch or unified designs. Multibranch architectures provide task-specific processing pathways, but their hard separation may restrict useful information exchange between localization and

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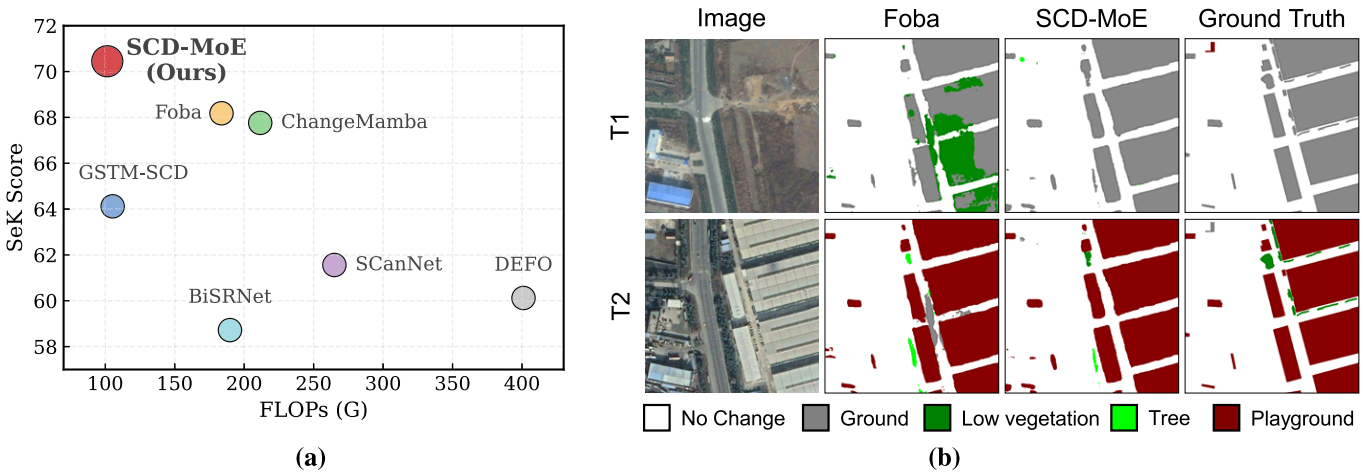


Fig. 1. (a) Our method achieves a favorable accuracy-FLOPs tradeoff compared with recent SCD methods. (b) Qualitative comparisons demonstrate that our method generates more precise change boundaries and better semantic consistency compared to the previous state-of-the-art.

recognition. Conversely, unified decoding strategies promote feature sharing, but a single shared representation may be insufficient to accommodate the different feature preferences of the two subtasks. This creates a tradeoff between task specialization and cross-task information sharing. Therefore, a more task-adaptive decoding paradigm is desirable, enabling the model to preserve useful shared information while generating suitable representations for different prediction objectives.

Taken together, these observations point to a common architectural requirement. SCD should support conditional specialization at two complementary levels: sample-conditioned interaction modeling and task-conditioned representation learning. Meeting this requirement entails maintaining multiple candidate feature transformations rather than relying on a single shared one, with their contributions adaptively determined according to the sample or task context. Mixture-of-experts (MoE)-style routing provides a natural mechanism for this purpose by organizing these transformations as specialized experts and using gating functions for context-dependent expert selection and aggregation.

Based on this insight, we propose SCD-MoE, a dynamic SCD framework that introduces tailored expert routing at both the bi-temporal interaction and task decoding stages. First, to achieve sample-conditioned interaction modeling, we introduce an interaction-adaptive MoE (IA-MoE) module. Rather than relying on a single shared fusion path, IA-MoE constructs difference-aware and semantic-preserving interaction streams as complementary information sources and integrates them through sample-adaptive expert routing. This enables the model to adjust its interaction behavior according to input-specific change characteristics. Second, to achieve task-conditioned representation learning, we develop a task-aware routing MoE (TAR-MoE) module. TAR-MoE employs task-specific gates over a shared expert pool, enabling change localization and semantic recognition to select different expert combinations while preserving controlled information sharing. The resulting task-aware features are then used to reconstruct time-specific semantic outputs and the binary change map. By combining interaction-level and task-level adaptive routing, SCD-MoE enables flexible spatiotemporal modeling with favorable accuracy and efficiency, as highlighted in Fig. 1.

Our contributions are summarized as follows.

1) We propose SCD-MoE, an efficient dynamic framework for remote sensing SCD that leverages adaptive expert routing to model sample-dependent bi-temporal interactions and task-specific feature preferences, offering a flexible paradigm for interpreting land-cover transitions.
2) We propose IA-MoE, which constructs difference-aware and semantic-preserving interaction streams and integrates them through sample-adaptive expert routing, enabling flexible bi-temporal interaction beyond a single shared fusion path.
3) We develop TAR-MoE, which introduces task-specific gates over a shared expert pool to model different feature preferences of change localization and semantic recognition, enabling controlled information sharing without rigid architectural separation.
4) Extensive experiments on three benchmarks demonstrate that SCD-MoE consistently outperforms existing methods both quantitatively and qualitatively, while achieving a favorable accuracy–efficiency tradeoff.

The remainder of this article is organized as follows. Section II reviews related work on remote sensing SCD and MoE. Section III presents the proposed method, including the overall SCD-MoE framework and detailed descriptions of the IA-MoE and TAR-MoE. Section IV reports the experimental settings, quantitative and qualitative comparisons, ablation studies, and further analyses. Section V discusses the limitations and future directions. Finally, Section VI concludes this article.

II. Related Work

A. Remote Sensing SCD

SCD aims to jointly localize changed regions and recognize their semantic transitions between bi-temporal remote sensing images. Compared with conventional change detection [25], [26], which only determines whether a change occurs, SCD

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requires fine-grained semantic interpretation at each temporal phase, making it a critical yet challenging component of high-level remote sensing image understanding [27].

To address these requirements, early SCD studies primarily relied on handcrafted change measures, such as change vector analysis [26]. With the development of deep learning, SCD has generally been formulated as a multitask problem involving semantic recognition and change localization. Existing methods have consequently focused on two related aspects: learning effective bi-temporal representations and jointly modeling change and semantic information. For bi-temporal representation learning, feature extractors have evolved from convolutional neural networks [6], [28], [29] to Transformers [20], [30] and state-space architectures [24], [31], [32], [33], progressively improving semantic representation and long-range contextual modeling. Beyond backbone design, various interaction mechanisms have been developed to capture temporal relations, ranging from feature concatenation [19], [20] to cross-attention [22], [23], [34] and gated fusion [24]. More specialized designs further model cross-temporal semantic correlations or spatiotemporal dependencies [20], [29], [31]. These approaches have effectively enhanced the capability to capture subtle discrepancies in bi-temporal remote sensing images. The joint modeling of change and semantic information has also been explored from complementary perspectives. In [22] and [32], semantic-change constraints are introduced to maintain consistency between bi-temporal semantic predictions and binary change results. At the architectural level, existing methods adopt either unified pipelines to reduce architectural redundancy or multibranch decoders with explicit feature interaction for information sharing [24], [35].

Despite these advances, existing SCD approaches still provide limited flexibility at both the bi-temporal interaction and task decoding levels. Most bi-temporal interaction strategies rely on a predefined set of transformations and an interaction pathway shared across samples. Although attention- or gating-based modules can generate input-dependent responses, the overall interaction formulation is generally fixed, making it difficult to accommodate diverse change patterns through different interaction behaviors. Second, existing methods do not sufficiently account for the distinct feature preferences of change localization and semantic recognition. The former relies more on discrepancy-sensitive and boundary-aware cues to identify changed regions, while the latter requires class-discriminative and region-consistent representations to preserve semantic consistency. Existing decoders typically adopt unified or separated multibranch structures, emphasizing either information sharing or task specialization but fixing this tradeoff through predefined pathways. Consequently, they still lack a flexible mechanism to preserve useful shared information while generating task-suitable representations for different prediction objectives. These limitations motivate SCD-MoE, which introduces conditional expert routing into both bi-temporal interaction and task decoding, enabling adaptive selection of feature transformations for different change patterns and prediction objectives.

B. Mixture of Experts

The MoE architecture was originally pioneered in [36] and [37] to decompose complex problem spaces into simpler subspaces handled by specialized subnetworks. A central idea of MoE is conditional computation, where a gating function dynamically selects or weights suitable experts according to the input or task [38]. In large-scale sparse MoE models, this mechanism is often used to expand model capacity without activating all parameters and has been widely explored in language modeling and vision tasks [39], [40], [41]. By routing only a subset of tokens to selected experts, these models improve the tradeoff between capacity and computational cost. In such settings, expert routing is associated with token-level dispatching, capacity scaling, and load-balancing for distributed computation. Therefore, expert utilization and routing balance are often central concerns in large-scale MoE systems.

Beyond large-scale capacity scaling, MoE has also been adopted as a compact routing mechanism for adaptive representation learning. In this line of work, a small set of experts is used as learnable candidate transformations, and the gate learns to selectively combine them according to input characteristics, domain variations, or task-specific requirements [42], [43], [44], [45]. Compared with dense modules using a single shared transformation, compact MoE-style routing provides a flexible way to model heterogeneous data distributions and different feature preferences without necessarily relying on a large number of experts. Such compact routing designs are particularly relevant when data heterogeneity or task diversity requires more flexible representation modeling.

In remote sensing image interpretation, recent studies have explored MoE across a range of applications. To cope with complex scene content and difficult category discrimination, expert-based architectures have been introduced for scene recognition [46], [47]. Multimodal remote sensing modeling has further adopted MoE to balance cross-modal knowledge shared with modality-specific representation learning and alleviate interference among heterogeneous observations [48], [49], [50]. For vision–language understanding, instruction-, task-, and granularity-aware routing has been used to specialize expert processing for different interpretation requirements [51], [52]. More closely related to our task, MoE has also been incorporated into optical-SAR change detection to accommodate distributional heterogeneity between the two modalities [53].

Collectively, these studies demonstrate the value of MoE for adaptive representation learning in remote sensing image interpretation. Remote sensing SCD also requires flexibility at both the sample and task levels: bi-temporal interaction should accommodate diverse change patterns, while decoding should account for the distinct feature preferences of change localization and semantic recognition. Compact MoE-style routing is well suited to this setting because it enables conditional selection among candidate feature transformations. Nevertheless, its application to these SCD-specific challenges remains insufficiently explored. Motivated by these characteristics and

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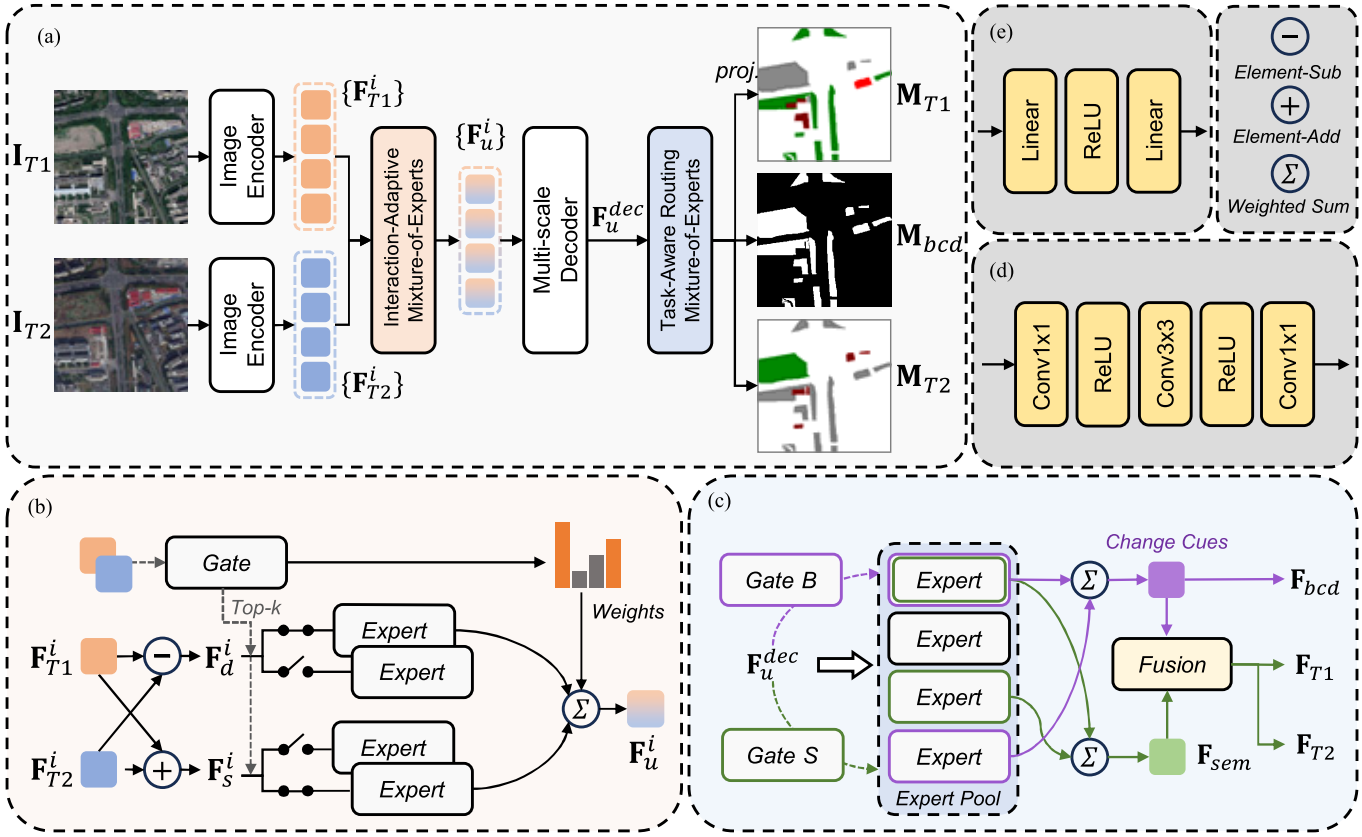


Fig. 2. (a) Overall framework of SCD-MoE, where bi-temporal features are extracted by a shared Siamese encoder, adaptively interacted by IA-MoE, decoded by a lightweight multiscale decoder, and further processed by TAR-MoE to generate the binary change map and temporal semantic maps. (b) IA-MoE at the interaction stage adaptively integrates difference-aware and semantic-preserving cues. (c) TAR-MoE at the decoding stage performs task-specific routing over a shared expert pool for task-oriented representation learning. (d) Architecture of the expert network. (e) Architecture of the gate network.

the remaining gap, we introduce SCD-MoE, a tailored framework that incorporates adaptive expert routing at both the bi-temporal interaction and decoding stages, providing a more flexible formulation for SCD.

III. Proposed Model

A. Overall Framework

We propose a dynamic SCD framework, termed SCD-MoE, for interpreting diverse and complex land-cover transitions. As illustrated in Fig. 2(a), the overall architecture follows a hierarchical encoder–decoder design, with two compact MoE modules introduced for adaptive bi-temporal interaction and task-aware feature learning. The data flow is summarized as follows.

1) Feature Extraction: Given the bi-temporal images $\mathbf{I}_{T1}, \mathbf{I}_{T2} \in \mathbb{R}^{H\times W\times 3}$, we first employ a weight-sharing Siamese encoder to extract hierarchical features. Let $\{\mathbf{F}_{T1}^i\}_{i=1}^L$ and $\{\mathbf{F}_{T2}^i\}_{i=1}^L$ denote the features extracted at the ith stage from the two temporal branches, where L indicates the number of scales.

2) Bi-Temporal Interaction: The extracted multiscale features are fed into the proposed IA-MoE for sample-adaptive bi-temporal interaction. IA-MoE constructs difference-aware and semantic-preserving interaction streams and integrates them through adaptive expert routing according to the input context. This process produces unified multiscale representations $\{\mathbf{F}_u^i\}_{i=1}^L$ that encode both change cues and semantic context.

3) Multiscale Decoding: Subsequently, a lightweight multiscale decoder is utilized to fuse these unified representations across different scales. The decoder generates a unified feature map, denoted as $\mathbf{F}_u^{dec}$, for subsequent change localization and semantic recognition.

4) Task-Aware Routing and Prediction: Finally, the decoded feature $\mathbf{F}_u^{dec}$ is processed by TAR-MoE. Through task-specific routing over a shared expert pool, TAR-MoE transforms the unified representation into task-oriented features, including the change-oriented feature $\mathbf{F}_{bcd}$ and time-specific semantic features $\mathbf{F}_{T1}$ and $\mathbf{F}_{T2}$. These features are then projected by convolutional heads to generate the final predictions: the binary change map $\mathbf{M}_{bcd}$ and the semantic segmentation maps $\mathbf{M}_{T1}$ and $\mathbf{M}_{T2}$.

B. Interaction-Adaptive MoE

Existing bi-temporal interaction strategies for SCD, such as feature differencing, concatenation with convolution, and attention-based fusion, have achieved promising progress in modeling cross-temporal dependencies. However, their overall interaction formulation or fusion topology is usually predefined and shared across samples. Although these modules can produce input-dependent responses, the interaction pathway

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itself is not explicitly adjusted for different change patterns or input samples. In real-world remote sensing scenarios, changes may exhibit diverse semantics, appearance variations, and occurrence frequencies. Therefore, a single shared interaction formulation may have limited flexibility in adapting to sample-specific change characteristics.

To address this issue, we propose the IA-MoE, which introduces sample-adaptive expert routing into bi-temporal interaction. Instead of relying on a single shared fusion path, IA-MoE provides multiple learnable interaction candidates and adaptively weights them according to the input context. This enables the network to flexibly combine complementary interaction cues for diverse change patterns.

As shown in Fig. 2(b), given the bi-temporal features $\mathbf{F}_{T1}^{i}, \mathbf{F}_{T2}^{i} \in \mathbb{R}^{H_i \times W_i \times C_i}$ at the ith stage, we first apply a lightweight 1×1 projection to map the bi-temporal features into a compact shared feature space for efficient expert processing. Then, at each encoder feature level, we concatenate the bi-temporal features and apply a global pooling operation to obtain a compact sample-level context descriptor $\mathbf{v}_{\text{gate}} \in \mathbb{R}^{D}$. This descriptor is fed into a lightweight gate network G to produce a scale-specific routing distribution over N experts

$$\mathbf{z} = \mathcal{G}\left(\mathbf{v}_{gate}\right), \quad \Omega = \text{TopK}\left(\mathbf{z}, K\right) \tag{1}$$

where Ω denotes the index set of the Top-K selected experts. The sparse routing weight for the jth expert is then computed by normalizing the logits within the selected expert set

$$g_j = \frac{\exp\left(z_j\right)}{\sum_{m \in \Omega} \exp\left(z_m\right)} \cdot \mathbb{I}\left(j \in \Omega\right) \tag{2}$$

where $\mathbb{I}(\cdot)$ is the indicator function. This formulation ensures that only the selected experts contribute to the output, while their contributions are normalized by the gate probabilities.

To provide complementary interaction cues, we explicitly construct two representations from the bi-temporal features: a difference feature $\mathbf{F}_{d}^{i} = \mathbf{F}_{T1}^{i} - \mathbf{F}_{T2}^{i}$, which emphasizes discrepancy-sensitive information, and a summation feature $\mathbf{F}_{s}^{i} = \mathbf{F}_{T1}^{i} + \mathbf{F}_{T2}^{i}$, which preserves semantic and structural context. These features are routed to two groups of experts with identical architectures but different interaction roles. Experts in the first group $\{E_1, \ldots, E_{N/2}\}$ process $\mathbf{F}_{d}^{i}$ to model difference-aware interactions, whereas experts in the second group $\{E_{N/2+1}, \ldots, E_N\}$ process $\mathbf{F}_{s}^{i}$ to model semantic-preserving interactions. Accordingly, the input to the jth expert is defined as

$$\mathbf{F}_{j}^{i} = \begin{cases} \mathbf{F}_{d}^{i}, & 1 \le j \le N/2 \\ \mathbf{F}_{s}^{i}, & N/2 < j \le N. \end{cases} \tag{3}$$

This design allows IA-MoE to selectively combine complementary transformations according to the input context.

Finally, the unified feature $\mathbf{F}_{u}^{i} \in \mathbb{R}^{H_i \times W_i \times C_i}$ is obtained by aggregating the outputs of the activated experts, with a residual term R implemented using a 1×1 convolution layer

$$\mathbf{F}_{u}^{i} = \sum_{j \in \Omega} g_j \cdot E_j\left(\mathbf{F}_{j}^{i}\right) + R\left(\left[\mathbf{F}_{T1}^{i}, \mathbf{F}_{T2}^{i}\right]\right) \tag{4}$$

where $[\cdot, \cdot]$ denotes channel-wise concatenation.

Overall, IA-MoE provides a flexible bi-temporal interaction module that adaptively integrates difference-aware and semantic-preserving transformations according to the input sample. Compared with a single shared interaction formulation, this design enables more adaptive feature interaction for SCD under diverse change scenarios.

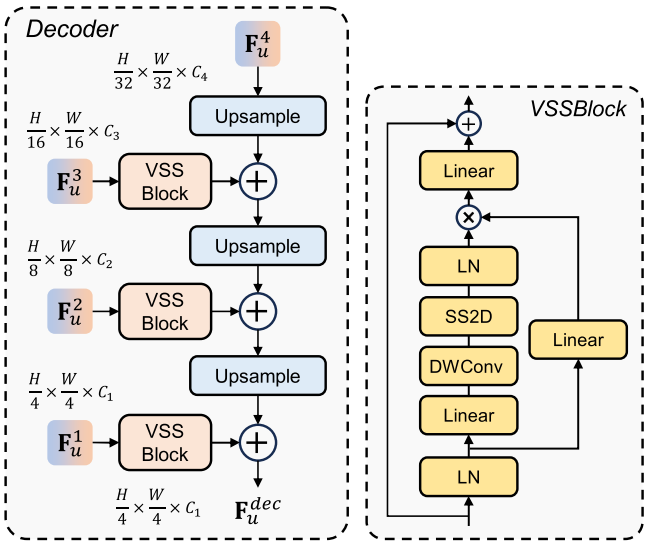


Fig. 3. Decoder architecture in SCD-MoE.

C. Multiscale Decoder

After IA-MoE performs bi-temporal interaction at each encoder stage, a multiscale decoder is employed to progressively aggregate the interacted features across different scales and produce a unified decoded representation for subsequent task-aware feature learning and prediction. Specifically, the decoder in SCD-MoE adopts a hierarchical top-down design to progressively refine multiscale bi-temporal features. It takes as input the fused representations $\{\mathbf{F}_{u}^{i}\}_{i=1}^{L}$ produced by the IA-MoE at $L = 4$ feature scales. The decoding process starts from the lowest-resolution feature and proceeds in a coarse-to-fine manner by gradually integrating finer spatial details through upsampling and feature fusion.

As illustrated in Fig. 3, at each decoding stage, the feature is first refined by a VSS block [54] to model spatial context and long-range dependencies. The refined feature is then fused with the upsampled representation from the previous coarser stage, enabling effective cross-scale information propagation.

D. Task-Aware Routing MoE

The decoding stage of SCD needs to support two related but different subtasks. The semantic segmentation branches aim to produce class-discriminative and region-consistent representations for recognizing land-cover categories at each temporal phase, whereas the binary change detection branch focuses more on discrepancy-sensitive and boundary-aware cues for localizing changed regions. A standard decoder that outputs a single unified feature representation may be insufficient to fully accommodate these different feature preferences. Alternatively, independent decoder branches can provide task-specific pathways, but their hard separation may limit useful

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information sharing between subtasks. This motivates a task-aware decoding mechanism that preserves useful cross-task information while generating task-suitable representations for different prediction objectives.

To address this issue, we propose the TAR-MoE, which introduces task-aware multigate routing over a shared expert pool. Instead of assigning separate experts to each subtask, TAR-MoE allows semantic recognition and change localization to learn different expert weightings from the same candidate transformation space. Specifically, as illustrated in Fig. 2(c), given the decoded feature $\mathbf{F}_u^{dec}$ output by the multistage decoder, we first employ a global pooling operation to extract a context vector $\mathbf{v}_{gate}^{dec}$. To capture task-specific requirements, we apply two distinct gate networks, $\mathcal{G}_{sem}$ and $\mathcal{G}_{bcd}$, to produce task-specific sample-level routing weights for the semantic segmentation and binary change detection branches, respectively. While the expert pool is shared across tasks, task-specific gate networks are employed to enable differentiated routing, allowing each task to assign different weights to the shared experts according to its own representation preference. The generation of these task-specific routing weights is formalized as

$$\begin{aligned} \mathbf{z}_{sem} &= \mathcal{G}_{sem}\left(\mathbf{v}_{gate}^{dec}\right), \quad \Omega_{sem} = \mathrm{TopK}\left(\mathbf{z}_{sem}, K\right) \\ \mathbf{z}_{bcd} &= \mathcal{G}_{bcd}\left(\mathbf{v}_{gate}^{dec}\right), \quad \Omega_{bcd} = \mathrm{TopK}\left(\mathbf{z}_{bcd}, K\right) \end{aligned} \tag{5}$$

where $\mathbf{z}_{sem}$ and $\mathbf{z}_{bcd}$ denote the routing logits, and Ω_{sem} and Ω_{bcd} denote the index sets of the Top-K selected experts for the semantic segmentation and binary change detection branches, respectively. The sparse routing weights are then obtained by normalizing the logits within each expert set

$$\begin{aligned} g_j^{sem} &= \frac{\exp\left(z_j^{sem}\right)}{\sum_{m\in\Omega_{sem}}\exp\left(z_m^{sem}\right)} \cdot \mathbb{I}\left(j \in \Omega_{sem}\right) \\ g_j^{bcd} &= \frac{\exp\left(z_j^{bcd}\right)}{\sum_{m\in\Omega_{bcd}}\exp\left(z_m^{bcd}\right)} \cdot \mathbb{I}\left(j \in \Omega_{bcd}\right) \end{aligned} \tag{6}$$

where $\mathbb{I}(\cdot)$ is the indicator function.

Through this strategy, TAR-MoE establishes a task-aware routing mechanism over a shared expert pool $\{E_1, \ldots, E_N\}$. Instead of relying on either rigidly separated branches or a single shared feature weighting, $\mathbf{g}_{sem}$ and $\mathbf{g}_{bcd}$ serve as task-specific weighting functions that assign different importance to the shared experts. In this way, the semantic and change detection branches can emphasize different feature transformations from the same candidate space while still preserving useful information sharing. Consequently, the semantic branch is encouraged to focus more on class-discriminative and region-consistent cues, whereas the change detection branch can emphasize discrepancy-sensitive and boundary-aware cues. This design provides a flexible way to generate task-suitable representations from the unified decoder feature.

Finally, the task-oriented features $\mathbf{F}_{sem}$ and $\mathbf{F}_{bcd}$ are obtained by weighted aggregation of the activated expert outputs and the residual term

$$\mathbf{F}_{sem} = \sum_{j\in\Omega_{sem}} g_j^{sem} \cdot E_j\left(\mathbf{F}_u^{dec}\right) + R\left(\mathbf{F}_u^{dec}\right)$$

$$\mathbf{F}_{bcd} = \sum_{j\in\Omega_{bcd}} g_j^{bcd} \cdot E_j\left(\mathbf{F}_u^{dec}\right) + R\left(\mathbf{F}_u^{dec}\right). \tag{7}$$

To allow temporal semantic prediction to incorporate change-aware context, we further fuse the semantic-oriented and change-oriented features to reconstruct the temporal representations for each phase

$$\begin{aligned} \mathbf{F}_{T1} &= H_F^{T1}\left(\mathbf{F}_{bcd}, \mathbf{F}_{sem}\right) \\ \mathbf{F}_{T2} &= H_F^{T2}\left(\mathbf{F}_{bcd}, \mathbf{F}_{sem}\right) \end{aligned} \tag{8}$$

where $H_F^{T1}(\cdot)$ and $H_F^{T2}(\cdot)$ denote two fusion modules implemented by two 1×1 convolutional layers with BatchNorm and ReLU. By fusing the change-oriented feature with the semantic-oriented feature, the temporal semantic branches can incorporate change-aware context during prediction.

Overall, TAR-MoE establishes a flexible, task-aware decoding mechanism. By leveraging multigate routing over a shared expert pool, it helps accommodate the different feature preferences of semantic recognition and change localization. This design extracts task-suitable representations while enabling controlled information exchange, which contributes to more accurate SCD.

1) Implementation of Experts and Gates: The expert blocks in IA-MoE and TAR-MoE adopt the same architecture, while the parameters of each expert are independent. Specifically, each expert is implemented as a 1×1 Conv-ReLU-3×3 Conv-ReLU-1×1 Conv structure with a 256-channel latent width, resulting in a 3×3 receptive field. The experts are independently initialized. For routing, the gate network extracts compact global descriptors by combining global average pooling and global max pooling, and then feeds them into a two-layer MLP to predict routing logits. Top-K selection and weight renormalization are subsequently applied to obtain sparse aggregation weights for expert fusion.

E. Training Objective

The proposed SCD-MoE is trained with a joint objective consisting of binary change detection supervision, SCD supervision, and temporal semantic consistency regularization. The overall loss function is defined as

$$\mathcal{L} = \mathcal{L}_{\mathrm{BCD}} + \mathcal{L}_{\mathrm{SCD}} + \lambda\mathcal{L}_{\mathrm{sim}} \tag{9}$$

where $\mathcal{L}_{\mathrm{BCD}}$ and $\mathcal{L}_{\mathrm{SCD}}$ supervise the binary change detection and SCD branches, respectively. $\mathcal{L}_{\mathrm{sim}}$ enforces semantic consistency between bi-temporal predictions in unchanged regions, and λ is a balancing coefficient, which is set to 0.5 in our experiments.

For the binary change detection branch, we adopt a hybrid loss consisting of the cross-entropy loss and the Lovasz-Softmax loss

$$\mathcal{L}_{\mathrm{BCD}} = \mathcal{L}_{\mathrm{ce}}\left(\mathbf{P}_b, \mathbf{G}_b\right) + 0.75\mathcal{L}_{\mathrm{lovasz}}\left(\mathbf{P}_b, \mathbf{G}_b\right) \tag{10}$$

where $\mathbf{P}_b$ and $\mathbf{G}_b$ denote the binary change prediction and the corresponding binary ground truth, respectively.

Similarly, the SCD branch is supervised by

$$\mathcal{L}_{\mathrm{SCD}} = \mathcal{L}_{\mathrm{ce}}\left(\mathbf{P}_s, \mathbf{G}_s\right) + 0.75\mathcal{L}_{\mathrm{lovasz}}\left(\mathbf{P}_s, \mathbf{G}_s\right) \tag{11}$$

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where $\mathbf{P}_s$ and $\mathbf{G}_s$ denote the semantic prediction and the corresponding semantic ground truth, respectively.

In addition, semantic predictions at two temporal phases are expected to remain consistent in unchanged regions. Therefore, following [24], we impose a similarity constraint between the semantic probability maps of T_1 and T_2. Let $\sigma(\cdot)$ denote the Softmax function. The similarity loss is formulated as

$$\mathcal{L}_{\text{sim}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \left\| \sigma\left(\mathbf{P}_{s1}^{(i)}\right) - \sigma\left(\mathbf{P}_{s2}^{(i)}\right) \right\|_2^2 \tag{12}$$

where $\mathbf{P}_{s1}$ and $\mathbf{P}_{s2}$ denote the semantic logits predicted for T_1 and T_2, respectively. $\mathcal{M}$ represents the set of unchanged pixels, which is derived from the binary change ground truth.

IV. Experiments

A. Experimental Settings

1) Dataset: We evaluate our method on three benchmark datasets spanning diverse spatial resolutions and land-cover categories: SECOND, JL1, and Landsat-SCD. The SECOND dataset [55] consists of 4662 pairs of 512 × 512 aerial images with spatial resolutions ranging from 0.5 to 3 m. It annotates semantic changes across six land-cover classes: nonvegetated ground surfaces, trees, low vegetation, water, buildings, and playgrounds. The JL1 dataset specifically targets farmland-type SCD. It comprises 6000 images with a size of 256 × 256 and a resolution of 0.75 m, covering five semantic categories: cropland, road, forest, building, and other. The Landsat-SCD [56] dataset is composed of 2425 image pairs with a spatial size of 416 × 416 pixels and a spatial resolution of 30 m. It involves four land-cover categories, including farmland, desert, building, and water.

2) Implementation Details: We adopt VMamba-Tiny [54] as the encoder, generating multiscale features at 1/4, 1/8, 1/16, and 1/32 resolutions with channel dimensions of 96, 192, 384, and 768, respectively. All models are optimized using the AdamW optimizer with a weight decay of 5×10^{-4} and an initial learning rate of 1×10^{-4}. To accommodate varying image resolutions and maximize GPU utilization, the batch sizes are set to 2, 12, and 8 for SECOND, JL1, and Landsat-SCD, respectively. The training process lasts for 800k iterations. For IA-MoE and TAR-MoE, we employ 4 experts and activate the Top-2. During training, a small amount of noise is injected into the gating logits to encourage exploration and stabilize expert routing.

3) Evaluation Metrics: For evaluation, we adopt three widely used metrics for SCD, including F_{scd}, mean intersection over union (mIoU), and the separated Kappa coefficient (SeK). Let $\mathbf{Q} = \{q_{ij}\}$ denote the confusion matrix, where q_{ij} is the number of pixels predicted as class i and labeled as class j. Class 0 denotes the unchanged category, and classes 1 to N denote changed semantic categories. The mIoU is computed as

$$IoU_{nc} = q_{00} \Big/ \left(\sum_{i=0}^{N} q_{i0} + \sum_{j=0}^{N} q_{0j} - q_{00} \right)$$

$$IoU_c = \sum_{i=1}^{N} \sum_{j=1}^{N} q_{ij} \Big/ \left(\sum_{i=0}^{N} \sum_{j=0}^{N} q_{ij} - q_{00} \right)$$

$$mIoU = \left(IoU_{nc} + IoU_c\right)/2. \tag{13}$$

To reduce the dominance of unchanged pixels, SeK is computed based on an adjusted confusion matrix $\hat{\mathbf{Q}} = \{\hat{q}_{ij}\}$, where $\hat{q}_{ij} = q_{ij}$ except that $\hat{q}_{00} = 0$

$$\rho = \sum_{i=1}^{N} \hat{q}_{ii} \Big/ \sum_{i=0}^{N} \sum_{j=0}^{N} \hat{q}_{ij}$$

$$\eta = \sum_{i=1}^{N} \left(\sum_{j=0}^{N} \hat{q}_{ij} \cdot \sum_{j=0}^{N} \hat{q}_{ji} \right) \Big/ \left(\sum_{i=0}^{N} \sum_{j=0}^{N} \hat{q}_{ij} \right)^2$$

$$SeK = e^{IoU_c - 1} \cdot (\rho - \eta) / (1 - \eta). \tag{14}$$

The F_{scd} score jointly considers the precision and recall of changed semantic categories

$$P_{scd} = \sum_{i=1}^{N} q_{ii} \Big/ \sum_{i=1}^{N} \sum_{j=0}^{N} q_{ij}$$

$$R_{scd} = \sum_{i=1}^{N} q_{ii} \Big/ \sum_{i=0}^{N} \sum_{j=1}^{N} q_{ij}$$

$$F_{scd} = 2 \cdot P_{scd} \cdot R_{scd} / \left(P_{scd} + R_{scd}\right). \tag{15}$$

Specifically, mIoU mainly reflects binary change localization performance, while F_{scd} jointly evaluates change localization and semantic correctness. SeK further mitigates the dominance of nonchange pixels by excluding their influence from the adjusted confusion matrix. The subscript C in SeK_C denotes the total number of transition categories.

B. Comparison With State-of-the-Art Methods

1) Quantitative Results: We compare our method with previous state-of-the-art SCD approaches, including SSCDl [29], BiSRNet [29], SCanNet [20], DEFO [35], CdSCNet [22], ChangeMamba [31], Foba [24], GSTM-SCD [32], and CHSCD [34]. In addition to accuracy metrics, we report the number of parameters and active FLOPs for all compared methods. For all methods, FLOPs are measured with a batch size of 1 using a single bi-temporal image pair, where each image has a spatial size of 512 × 512. For SCD-MoE, the reported parameter count includes all experts in the expert pools, including those not selected for a given input. In contrast, the reported active FLOPs correspond to one forward pass under Top-2 routing, where only the routed Top-2 experts are executed in each MoE module, and the unselected experts incur no expert-computation cost. The quantitative evaluation results are reported in Table I and lead to the following observations.

First, SCD-MoE outperforms the strongest competing methods by 0.62%, 2.27%, and 5.04% in terms of SeK on the SECOND, JL1, and Landsat-SCD datasets, respectively, while also delivering consistent improvements in mIoU. These improvements in both change localization and semantic transition recognition can be mainly attributed to the complementary MoE designs: IA-MoE provides flexible, input-adaptive interaction pathways that enhance bi-temporal representation learning, while TAR-MoE enables the localization and recognition branches to emphasize their respective task-relevant

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TABLE I

QUANTITATIVE COMPARISON. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SECOND-BEST ARE UNDERLINED

Method	SECOND			JL1			Landsat-SCD			Params	GFLOPs
	F_{scd}	mIoU	SeK_{37}	F_{scd}	mIoU	SeK_{26}	F_{scd}	mIoU	SeK_{17}		
SSCDl [29]	53.90	73.08	21.62	83.43	84.38	55.35	81.77	83.47	43.73	23.31	189.76
BiSRNet [29]	54.57	73.27	22.16	84.86	85.96	58.72	83.84	85.35	48.33	23.38	190.30
SCanNet [20]	54.77	73.21	22.15	86.73	86.83	61.56	88.30	88.77	57.69	27.90	264.95
DEFO [35]	54.17	73.62	22.32	85.12	86.53	60.12	87.83	89.33	58.50	26.02	401.09
CdSCNet [22]	55.80	72.99	22.56	83.50	83.20	53.82	85.56	88.25	54.79	27.86	521.92
ChangeMamba [31]	54.93	73.28	22.54	89.67	89.05	67.76	86.03	88.37	54.86	89.99	211.55
Foba [24]	57.61	74.17	24.29	89.70	89.32	68.18	87.91	88.93	57.65	86.32	183.68
GSTM-SCD [32]	55.87	73.25	22.79	87.68	87.86	64.12	86.51	87.98	55.57	52.77	105.31
CHSCD [34]	54.87	73.41	22.47	83.16	83.90	54.51	86.87	88.65	56.48	26.87	407.36
SCD-MoE	**57.75**	**74.32**	**24.91**	**90.73**	**90.15**	**70.45**	**90.15**	**90.73**	**63.54**	41.32	101.44

TABLE II

PER-CLASS F_1 PERFORMANCE COMPARISON ON THE SECOND DATASET AMONG REPRESENTATIVE METHODS. THE "PROPORTION" ROW INDICATES THE PERCENTAGE OF PIXELS FOR EACH CLASS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SECOND-BEST ARE UNDERLINED

Method	Water	Ground	Low Veg.	Tree	Building	Playground
Proportion	22.72	37.41	5.95	1.24	32.05	0.63
SCanNet	52.34	61.87	41.47	50.53	75.85	72.59
CdSCNet	53.93	62.61	41.99	54.78	76.53	74.57
ChangeMamba	52.77	62.49	42.38	54.59	76.56	71.70
Foba	52.87	**65.41**	42.71	53.64	**77.43**	74.20
GSTM-SCD	53.39	63.37	40.01	54.53	76.96	70.92
CHSCD	54.07	62.73	38.82	53.16	75.47	71.08
Ours	**54.39**	65.23	**43.93**	**56.55**	77.36	**75.07**

cues. Second, SCD-MoE exhibits a favorable tradeoff between performance and efficiency. Despite achieving superior accuracy, our default SCD-MoE with VMamba-Tiny uses only 48% of the parameters of Foba [24] under its official VMamba-Base configuration and achieves the lowest active FLOPs among the compared methods. This efficiency is mainly attributed to the compact design of SCD-MoE. IA-MoE provides a compact and flexible alternative to conventional bi-temporal interaction modules, while TAR-MoE improves parameter efficiency by sharing the expert pool across tasks instead of relying on fully independent task-specific decoders. It is worth noting that the SECOND dataset poses a more challenging evaluation setting due to its finer-grained annotations and a larger number of semantic categories. Under such increased semantic complexity, SCD-MoE still achieves consistent improvements over existing methods, attaining the best SeK score of 24.91, which further demonstrates its effectiveness on fine-grained SCD scenarios.

In Table II, we further compare the per-class performance of SCD-MoE against representative top-performing methods on the SECOND dataset. For dominant categories with high pixel proportions, such as ground and building, our method achieves performance comparable to the strongest competing method, Foba [24]. Meanwhile, SCD-MoE obtains clear improvements on water, which also accounts for a large portion of the dataset. More importantly, our method shows consistent advantages on less frequent categories, such as low vegetation and tree, improving the F_1 scores from 42.71 to 43.93 and from 54.78 to 56.55, respectively. These results suggest that the proposed sample-adaptive interaction is beneficial for

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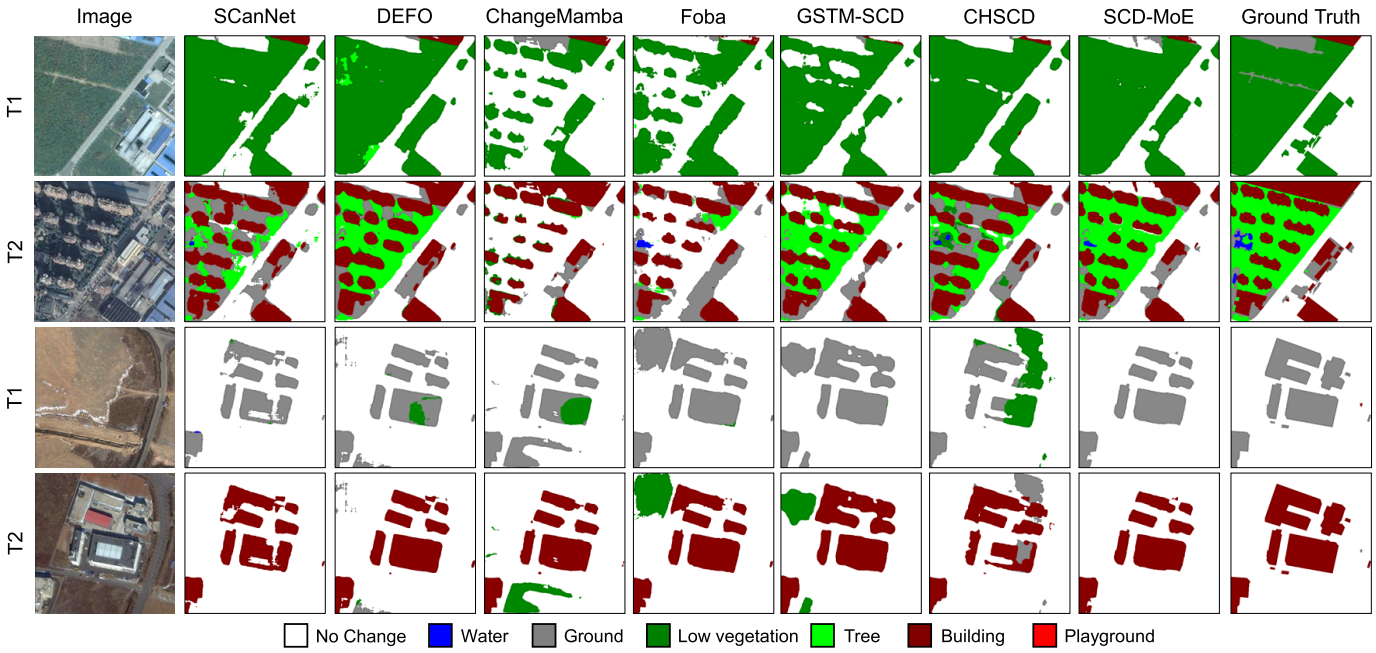


Fig. 4. Qualitative comparison of different methods on representative samples from the SECOND dataset.

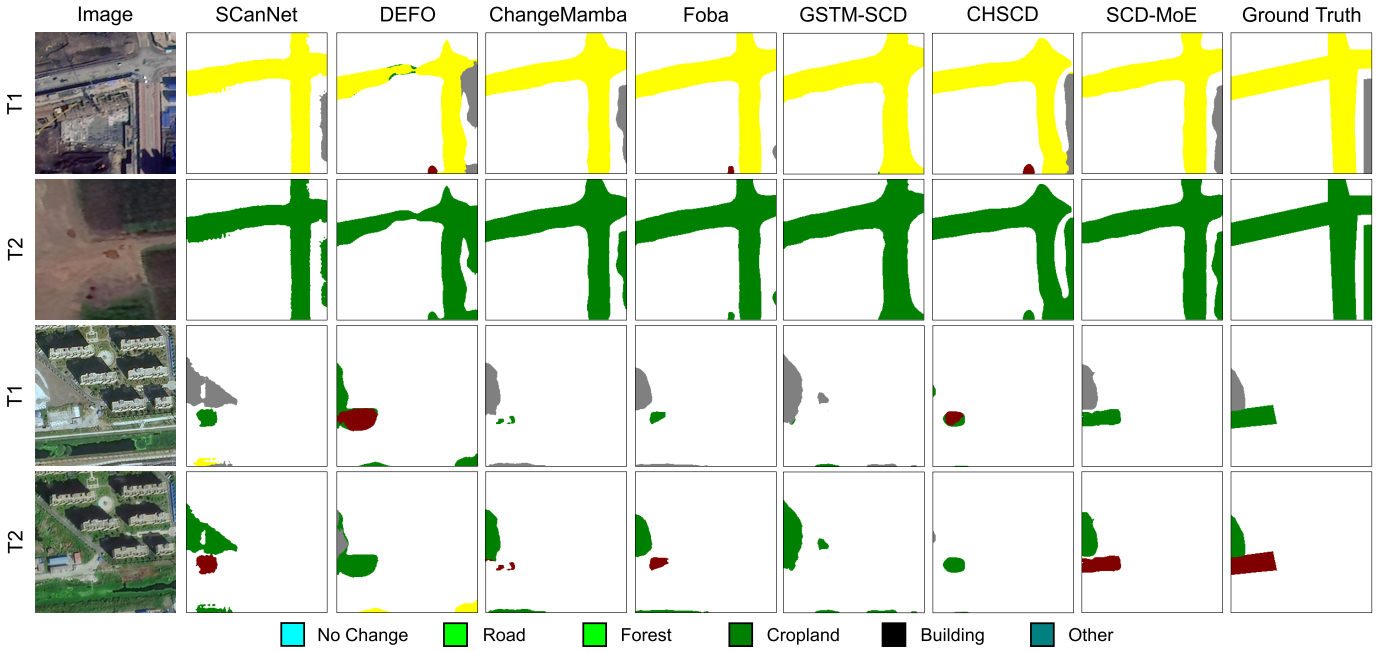


Fig. 5. Qualitative comparison of different methods on representative samples from the JL1 dataset.

categories with fewer samples, possibly because it provides more flexible bi-temporal modeling than a single shared interaction formulation.

2) Qualitative Results: Figs. 4–6 present qualitative comparisons between SCD-MoE and several competitive methods on the SECOND, JL1, and Landsat-SCD datasets, respectively. Across these datasets with different spatial resolutions and land-cover distributions, SCD-MoE consistently produces more accurate and spatially coherent predictions. Overall, two major advantages can be observed.

First, SCD-MoE demonstrates robust structural integrity and spatial consistency. As shown in the first scene of Fig. 4, our method accurately segments extensive tree regions with smoother boundaries and more compact structures, while competing methods tend to produce noisy and fragmented outputs. In the first scene of Fig. 5, SCD-MoE more accurately captures the narrow road branch on the right side, whereas competing methods either miss this fine structure or produce small spurious responses. For the Landsat-SCD dataset, despite the limited spatial details, our method more reliably identifies Water bodies and maintains coherent spatial layouts, as shown in the first scene of Fig. 6.

Second, SCD-MoE exhibits superior semantic discriminability in complex land-cover transitions. As shown in

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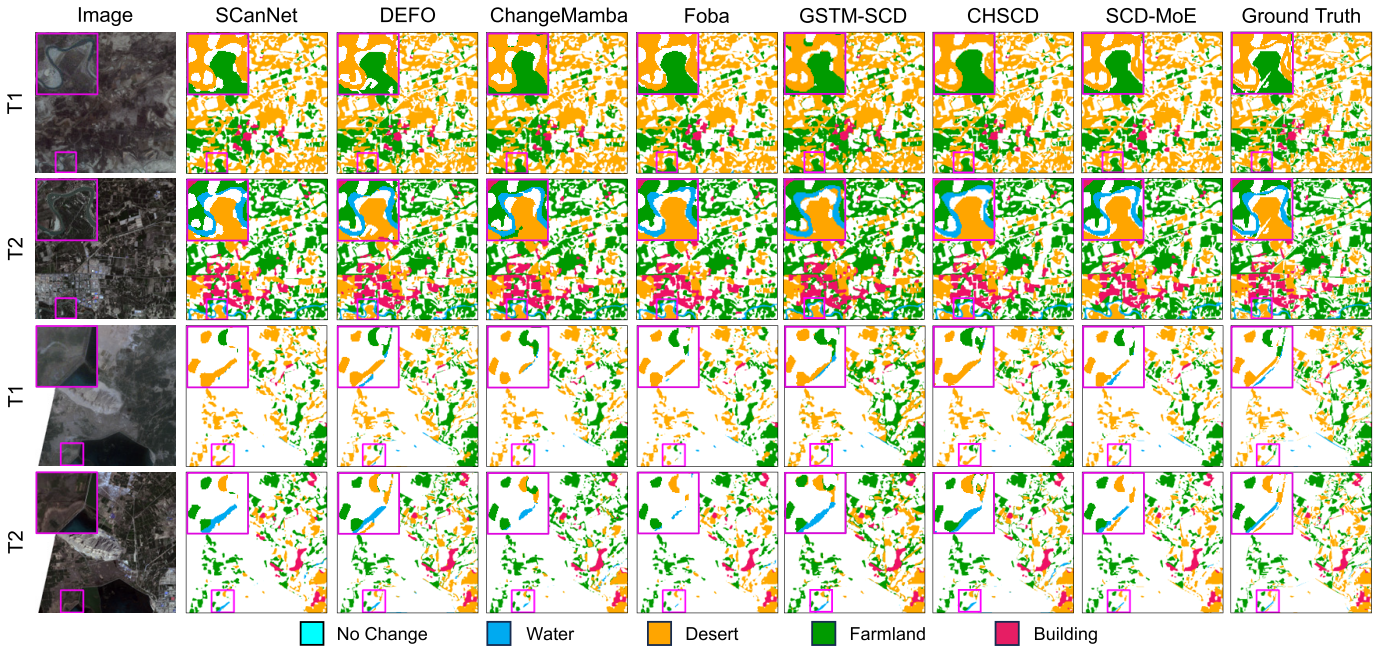


Fig. 6. Qualitative comparison of different methods on representative samples from the Landsat-SCD dataset. Due to the relatively low spatial resolution of Landsat-SCD, zoomed-in views are provided for better visualization.

the second scene of Fig. 4, our method effectively captures building changes and suppresses pseudo changes that commonly appear in the predictions of competing methods. In the second scene of Fig. 5, SCD-MoE accurately identifies cropland-to-building transitions, whereas competing methods either generate fragmented predictions or fail to capture this change clearly. Moreover, in the second scene of Fig. 6, SCD-MoE better captures subtle desert-water transitions in the highlighted regions, which are partially missed or less precisely delineated by other methods. These qualitative comparisons demonstrate that SCD-MoE not only improves semantic category discrimination but also enhances the spatial consistency of change predictions across different scenarios. These observations are consistent with the designs of the proposed modules: IA-MoE adaptively balances discrepancy-sensitive and semantic-preserving interactions according to the input, while TAR-MoE generates task-oriented representations that emphasize the information preferred by change localization and semantic recognition, respectively.

C. Ablation and Analysis

1) Ablation on IA-MoE: We first analyze the key design choices of IA-MoE, as shown in the upper part of Table III. The baseline "w/o IA-MoE" replaces the proposed interaction module with a 1×1 convolution over the concatenated bi-temporal features. "Standard MoE" directly feeds the concatenated bi-temporal features into a generic MoE module without the explicit difference/summation decomposition. To further disentangle the effect of the SCD-specific decomposition from expert routing, we additionally evaluate "Diff/Sum + Conv," where the difference feature $\mathbf{F}_{T1} - \mathbf{F}_{T2}$ and the summation feature $\mathbf{F}_{T1} + \mathbf{F}_{T2}$ are concatenated and projected by a 1×1 convolution without MoE routing. Moreover, we

TABLE III
ABLATION STUDY ON IA-MOE AND TAR-MOE. THE PERFORMANCE ON THE SECOND DATASET IS REPORTED

Module	Method	F_{scd}	mIoU	SeK_{37}
IA-MoE	w/o IA-MoE	57.37	73.98	24.10
	Standard MoE	57.02	73.64	23.93
	Diff/Sum + Conv	57.34	73.94	24.02
	MoE w/o Routing	57.35	74.09	24.32
	Ours	**57.75**	**74.32**	**24.91**
TAR-MoE	w/o TAR-MoE	56.93	74.20	24.40
	Shared Gating	56.48	**74.44**	24.49
	Separate Expert Pools	56.97	74.23	24.24
	Ours	**57.75**	74.32	**24.91**

introduce the "MoE w/o Routing" variant to evaluate the contribution of adaptive routing, where the expert branches are preserved, but their outputs are aggregated with fixed weights instead of sample-adaptive routing.

As can be seen from the results, the generic "Standard MoE" performs worse than the simple convolutional baseline, indicating that naively applying MoE to concatenated bi-temporal features is insufficient for SCD. Meanwhile, "Diff/Sum + Conv" obtains an SeK score of 24.02%, which is comparable to but slightly lower than "w/o IA-MoE." This result shows that the explicit difference/summation decomposition alone does not account for the performance gain. The variant "MoE w/o Routing" further improves SeK to 24.32%, indicating that the increased expert capacity can provide limited additional representation capability. However, it still underperforms the full IA-MoE by 0.59 percentage points in SeK. In contrast, the full IA-MoE achieves the best performance, suggesting

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TABLE IV
ROUTING PREFERENCE OF IA-MOE ACROSS DIFFERENT DATASETS

Dataset	Diff.	Sem.
SECOND	38.91	61.09
JL1	48.92	51.08
Landsat	71.04	28.96

TABLE V
TASK-SPECIFIC GATE DISTRIBUTION ANALYSIS OF TAR-MOE

Dataset	Gate Cos. Similarity	Gate L1 Distance
SECOND	0.238	0.392
JL1	0.509	0.313
Landsat-SCD	0.376	0.324

that IA-MoE benefits from the combination of SCD-specific interaction priors and adaptive expert routing, rather than from either component alone.

2) Ablation on TAR-MoE: We then evaluate the effectiveness of TAR-MoE in the lower part of Table III. The variant "w/o TAR-MoE" directly feeds the decoded unified feature map $\mathbf{F}_u^{dec}$ to the prediction heads, without task-specific feature learning. "Shared Gating" uses a single shared gate network for both the semantic segmentation and binary change detection branches, while keeping the same expert pool. "Separate Pools" assigns an independent gate and a complete expert pool to each branch, with each expert pool having the same architecture as the shared pool in TAR-MoE.

As can be seen from the results, removing TAR-MoE leads to clear performance degradation, especially in F_{scd} and SeK, demonstrating the importance of task-aware representation learning at the decoding stage. Interestingly, "Shared Gating" obtains a marginally higher mIoU, but its F_{scd} and SeK are clearly lower. Since mIoU mainly reflects binary change localization and does not explicitly evaluate semantic transition correctness, this result suggests that a shared routing strategy may slightly favor coarse change localization. However, it significantly decreases F_{scd} and SeK, indicating degraded semantic change consistency. Meanwhile, the "Separate Pools" variant introduces more parameters but provides no consistent improvement over the model without TAR-MoE, indicating that completely separating the two branches restricts the interaction and sharing of complementary semantic and change information. In contrast, the proposed TAR-MoE achieves a better balance between binary localization and semantic recognition through task-specific routing, which is more aligned with the objective of SCD.

3) IA-MoE Routing Behavior Analysis: We further analyze the routing preferences of IA-MoE across datasets. Specifically, we report the activation frequency of difference-aware experts and semantic-preserving experts across all feature levels. As shown in Table IV, IA-MoE exhibits noticeably different routing preferences on different datasets. On Landsat-SCD, the model selects difference-aware experts more frequently (71.04%), suggesting that the interaction module tends to emphasize discrepancy cues under coarse-resolution imagery. This behavior is consistent with the limited spatial details in Landsat images, where semantic structures are less fine-grained and change-related differences become more important for prediction. In contrast, on SECOND, IA-MoE favors semantic-preserving experts (61.09%), which is consistent with its finer spatial resolution and more diverse semantic categories, requiring richer semantic context for land-cover discrimination. These observations provide evidence that IA-MoE learns dataset-dependent routing preferences between difference-aware and semantic-preserving interactions.

4) Task-Specific Routing and Feature Analysis: To better understand the task-aware behavior of TAR-MoE, we analyze the gate distributions of the binary change detection branch and the semantic segmentation branch. As shown in Table V, we report two routing-level metrics. Gate Cos. Similarity measures the cosine similarity between the CD and semantic gate probability distributions, while Gate L1 Distance measures the average absolute difference between their expert weights. Let $\mathbf{g}_{\text{sem}}, \mathbf{g}_{\text{bcd}} \in \mathbb{R}^N$ denote the routing-weight vectors of the semantic recognition and change detection branches for a test sample, respectively, where N is the number of experts. The Gate Cosine Similarity and Gate L_1 Distance are defined as

$$S_{\cos} = \frac{\mathbf{g}_{\text{sem}}^{\top}\mathbf{g}_{\text{bcd}}}{\|\mathbf{g}_{\text{sem}}\|_2 \, \|\mathbf{g}_{\text{bcd}}\|_2}, \quad D_{L_1} = \|\mathbf{g}_{\text{sem}} - \mathbf{g}_{\text{bcd}}\|_1 . \tag{16}$$

The values reported in Table V are obtained by averaging the metrics over all test samples.

Across all three datasets, the Gate Cos. Similarity values are clearly below 1.0, and the Gate L1 Distance remains above 0.31. This indicates that the CD and semantic branches assign different probability weights to the shared expert pool, rather than using identical expert weighting. These observed routing differences suggest that TAR-MoE learns task-specific preferences over shared feature experts for change localization and semantic recognition.

Complementarily, as illustrated in the left part of Fig. 7, the learned feature representations exhibit task-oriented characteristics. The change-specific feature $\mathbf{F}_{cd}$ tends to highlight structural boundaries, providing cues for change localization. In contrast, the semantic-specific feature $\mathbf{F}_{sem}$ shows stronger region-level consistency and activates object interiors, which is beneficial for semantic recognition. This qualitative observation is consistent with the task-specific routing shown in the right part of Fig. 7, where $\mathcal{G}_{bcd}$ and $\mathcal{G}_{sem}$ assign different weights to the shared experts. Together, these results suggest that TAR-MoE learns task-aware routing and representation preferences, helping balance change-sensitive and semantic-preserving information in SCD.

5) Sensitivity Analysis: We conduct a sensitivity analysis on the total number of experts (E) and the number of activated experts (K) for both IA-MoE and TAR-MoE. As shown in Fig. 8, the configuration $E = 4, K = 2$ achieves a favorable tradeoff between performance and computational cost, as reflected by the corresponding Params and FLOPs bars.

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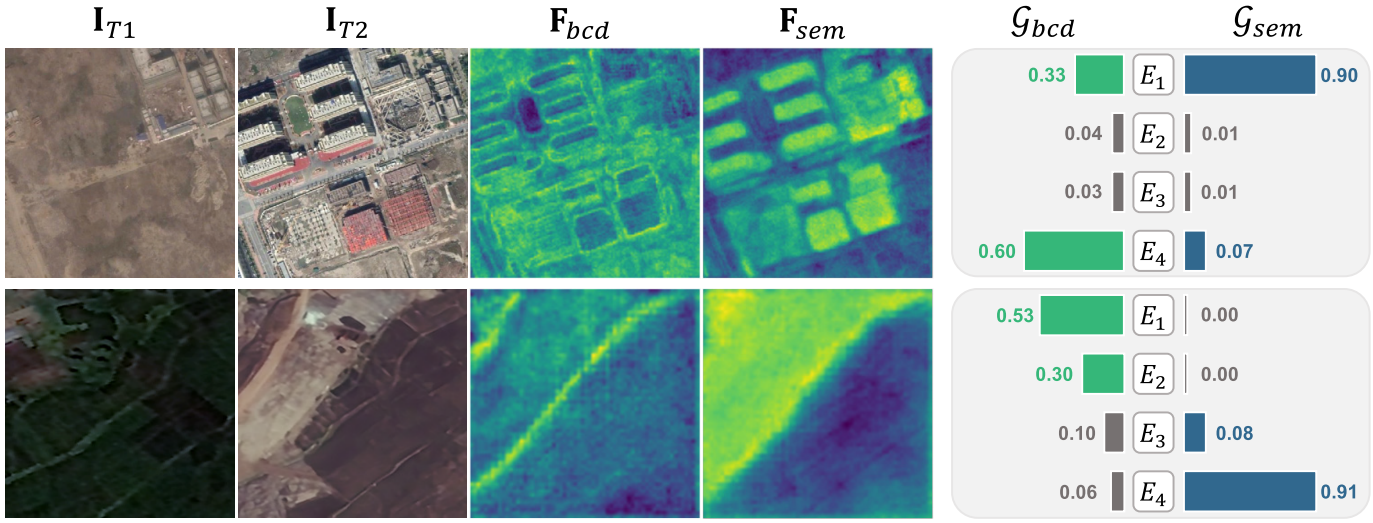


Fig. 7. Left: visualization of the task-oriented features in TAR-MoE. Right: distribution of task-specific gating weights.

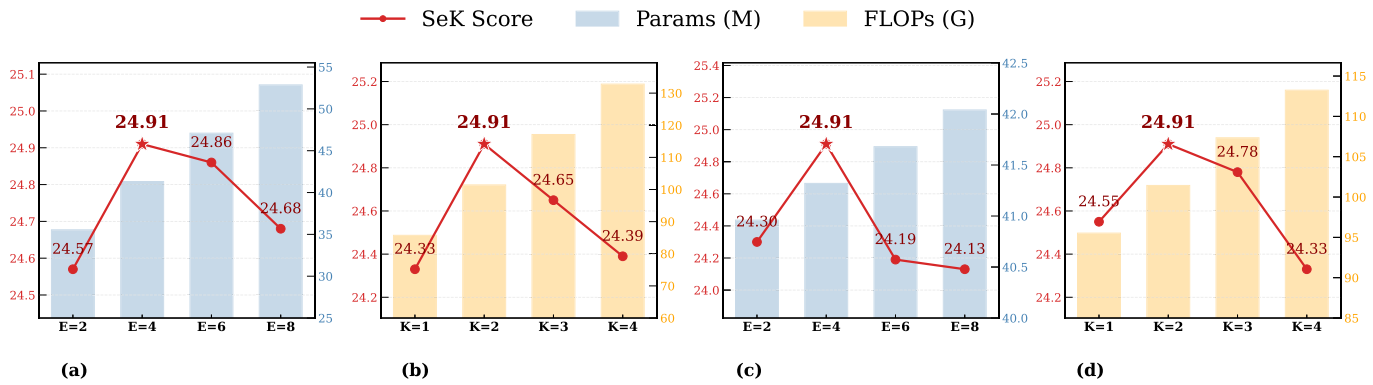


Fig. 8. Impact of expert numbers E and Top-K settings on performance and efficiency. When varying E, we activate the Top-2 experts; when varying Top-K, we fix $E = 4$. (a) IA-MoE: impact of E. (b) IA-MoE: impact of K. (c) TAR-MoE: impact of E. (d) TAR-MoE: impact of K.

For IA-MoE, increasing E beyond 4 does not bring further improvement and even leads to slight performance degradation. This suggests that the interaction modeling capacity may become saturated once sufficient expert transformations are provided, and an overly large expert pool may introduce redundant or less effective routing choices. For TAR-MoE, increasing E also results in degraded performance, indicating that a compact expert pool is more suitable for modeling task-specific feature preferences. Regarding routing sparsity, when fixing $E = 4$, activating all experts ($K = 4$) consistently underperforms sparse routing ($K = 2$) in both modules, despite involving more active parameters and higher computational cost. These observations support the importance of compact sparse routing in SCD-MoE, rather than simply increasing the number of experts or active parameters.

To better understand why increasing E or K does not consistently yield further gains, we analyze the concentration of routing preferences using the Top-K Routing Mass [57], [58]. As both IA-MoE and TAR-MoE exhibit a consistent sensitivity pattern, we use the former—whose routing is repeated across feature levels and thus yields broader routing observations—as a representative case for analysis. Specifically, we collect the routing logits generated by the IA-MoE gates at all feature levels and compute

$$\mathcal{M}_K = \sum_{j \in \Omega} \frac{\exp(z_j)}{\sum_{n=1}^{N} \exp(z_n)} \tag{17}$$

where N denotes the total number of experts. $\mathcal{M}_K$ measures the proportion of routing probability retained by the Top-K selected experts. A larger value indicates that the selected experts capture a greater proportion of the routing preference.

As shown in Fig. 9(a), with $E = 4$, Top-1 routing retains only a limited proportion of the routing probability, whereas Top-2 routing already retains nearly all of it. This indicates that the routing preference is effectively captured by two activated experts. Further increasing K involves more expert transformations in each routing decision, which may weaken sparse conditional computation and introduce redundant or less complementary interactions. In Fig. 9(b), the Routing Mass decreases as E increases under fixed Top-2 routing, indicating that the routing probability is distributed across more candidate experts while only two are selected. Moreover, under the same training data and fixed K, a larger expert pool receives fewer

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TABLE VI

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART MAMBA-BASED METHODS ON THE SECOND DATASET. BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST, RESPECTIVELY

Method	Backbone	F_{scd}	mIoU	SeK_{37}	Params	GFLOPs
ChangeMamba [31]	VMamba-Tiny	54.50	73.21	22.23	21.51	73.42
	VMamba-Base	54.93	73.28	22.54	89.99	211.55
Foba [24]	VMamba-Tiny	56.86	74.08	23.92	30.04	59.42
	VMamba-Base	57.61	74.17	24.29	86.32	183.68
Ours	VMamba-Tiny	<u>57.75</u>	<u>74.32</u>	**24.91**	41.32	101.44
	VMamba-Base	**57.80**	**74.47**	<u>24.75</u>	96.50	249.28

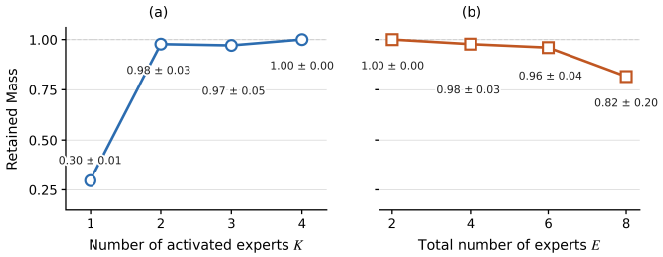


Fig. 9. Top-K routing mass of IA-MoE under varying expert configurations. (a) Fixed $E = 4$. (b) Fixed $K = 2$.

TABLE VII

INFERENCE SPEED COMPARISON ON THE SECOND DATASET

Method	SeK_{37}	FPS
SSCDl [29]	21.62	**152.12**
BiSRNet [29]	22.16	126.87
SCanNet [20]	22.15	74.62
DEFO [35]	22.32	65.50
CdSCNet [22]	22.56	67.73
ChangeMamba [31]	22.54	15.08
Foba [24]	24.29	28.61
GSTM-SCD [32]	22.79	17.56
CHSCD [34]	22.47	42.33
SCD-MoE	**24.91**	49.77

routing assignments per expert on average. These routing behaviors help explain the saturation or degradation observed in Fig. 8 and support the compact $E = 4$, $K = 2$ configuration.

6) Backbone Scaling Analysis: To further evaluate the accuracy–efficiency tradeoff of SCD-MoE under different backbone capacities, we compare different Mamba-based methods with VMamba-Tiny and VMamba-Base backbones. As presented in Table VI, SCD-MoE consistently achieves superior performance compared with existing Mamba-based SCD methods [24], [31] across backbone scales. Notably, under the VMamba-Tiny setting, SCD-MoE achieves an SeK of 24.91, outperforming Foba with VMamba-Base (24.29) while reducing the computational cost by nearly 45%. This result suggests that the proposed interaction- and task-level routing design can provide an effective accuracy–efficiency tradeoff without relying solely on larger backbone capacity.

We also observe that scaling the backbone from VMamba-Tiny to VMamba-Base does not bring consistent and significant gains for SCD-MoE. Although F_{scd} and mIoU are slightly improved, SeK decreases from 24.91 to 24.75, while the parameters and FLOPs increase substantially. This suggests that simply increasing backbone capacity is not necessarily beneficial under our current design. One possible reason is that the proposed MoE modules already provide interaction- and task-aware feature transformations, thereby reducing the marginal benefit of using a stronger backbone alone. In other words, the performance of SCD-MoE depends not only on stronger feature extraction, but also on how bi-temporal and task-specific information is adaptively routed and transformed. Therefore, the VMamba-Tiny setting provides a more favorable accuracy–efficiency tradeoff. In contrast, competing methods benefit more consistently from increased backbone capacity, indicating that their performance may depend more heavily on stronger feature extraction. These observations suggest that SCD-MoE can effectively use compact backbone features through adaptive routing, offering an efficient alternative to straightforward backbone scaling.

7) Runtime Analysis: To further evaluate the practical inference efficiency of different SCD models, we measure the inference speed of all methods in terms of frames per second (FPS). All experiments are conducted on a workstation equipped with a single NVIDIA GeForce RTX 4090 GPU and an Intel Core i9-13900K CPU, using a batch size of 1 and an input resolution of 512 × 512. We also report the SeK score on the SECOND dataset to reflect the accuracy–runtime tradeoff.

As shown in Table VII, SCD-MoE achieves the highest SeK score among the compared methods while maintaining a practical inference speed. Although SCD-MoE has the lowest GFLOPs in Table I, this theoretical efficiency does not directly translate into the highest FPS. This is because FLOPs mainly quantify arithmetic operations, whereas actual inference speed is also affected by practical execution factors, such as memory access overhead and routing overhead. In SCD-MoE, the MoE modules selectively activate different experts according to the input features, enabling adaptive interaction

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TABLE VIII
BACKBONE COMPARISON RESULTS ON THE SECOND DATASET

Backbone	MoE	F_{scd}	mIoU	SeK_{37}
ResNet-50	w/o	52.55	71.63	19.31
ResNet-50	w/	53.75 (+1.20)	72.20 (+0.57)	20.88 (+1.57)
MiT-B0	w/o	54.68	72.85	21.35
MiT-B0	w/	55.05 (+0.37)	73.47 (+0.62)	22.24 (+0.89)
DINOv3	w/o	55.46	73.38	22.64
DINOv3	w/	56.90 (+1.44)	74.78 (+1.40)	24.30 (+1.66)

modeling and task-oriented feature learning. This routing design improves accuracy by introducing input-conditioned flexibility, but the associated Top-K selection and dynamic expert dispatch inevitably introduce additional execution overhead. Notably, compared with recent high-performing Mamba-based methods, including Foba, GSTM-SCD, and ChangeMamba, SCD-MoE still achieves clear advantages in both accuracy and inference speed. These results suggest that the additional routing cost is effectively compensated by the compact expert design and improved representation capability, allowing SCD-MoE to provide a favorable accuracy–runtime tradeoff, especially among high-performing SCD models.

8) Cross-Backbone Generalization Analysis: To further validate the effectiveness and backbone generalizability of the proposed MoE modules, we conduct backbone ablations on the SECOND dataset using three representative encoders from different architectural families: ResNet-50 [59] as a CNN-based encoder, MiT-B0 [60] as a hierarchical Transformer encoder, and DINOv3 ViT-L/16 [61] as a foundation-model encoder. For each backbone, we compare models with and without the proposed MoE modules while keeping the remaining network components and training settings unchanged. As shown in Table VIII, introducing the proposed MoE modules consistently improves performance across all three encoders, yielding SeK gains of 1.57, 0.89, and 1.66 percentage points for ResNet-50, MiT-B0, and DINOv3 ViT-L/16, respectively. These results further demonstrate that the effectiveness of the proposed adaptive routing mechanisms is not restricted to VMamba and can generalize across different encoder architectures.

V. FURTHER ANALYSIS AND LIMITATIONS

A. Analysis of Routing Behavior

The routing behavior of SCD-MoE may differ from that of large-scale sparse MoE architectures. In such architectures, load balancing is often introduced to support scalability with a large number of experts. SCD-MoE instead employs a compact expert pool, in which each expert represents a learnable transformation rather than a distributed capacity block. Under this setting, some degree of nonuniform expert utilization can be naturally expected, since different feature representations may favor different interaction transformations according to the change patterns, semantic categories, and scene complexities in remote sensing images.

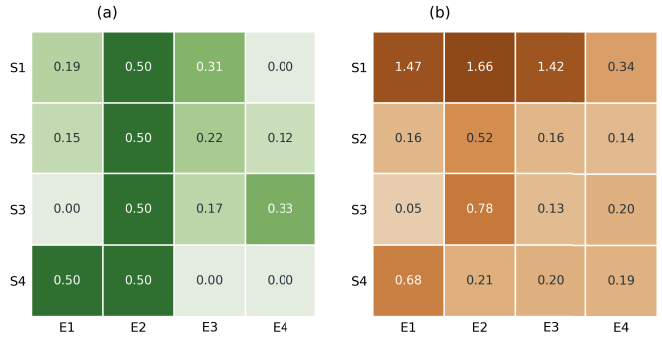


Fig. 10. Routing behavior analysis of IA-MoE. (a) Expert utilization across different feature scales. (b) Standard deviation of gating logits across sampled test instances. Here, S_1–S_4 denote the four encoder feature levels, and E_1–E_4 denote the corresponding experts at each level.

To better understand the learned routing behavior, we conduct additional analysis on the Landsat-SCD test set. Specifically, expert utilization is calculated as the proportion of all Top-K routing assignments assigned to each expert. In addition, the variation of gating logits across test instances is measured to examine whether the routing function remains input-dependent. As shown in Fig. 10, different feature scales exhibit distinct expert activation patterns. Some feature scales activate multiple experts with different utilization ratios, while the gating logits show nonnegligible variations. These observations indicate that the learned routing strategy is not reduced to a fixed expert combination, but adapts according to feature hierarchy and input characteristics. We do observe that some feature scales exhibit more concentrated routing patterns. However, this phenomenon does not necessarily indicate that IA-MoE degenerates into a fixed interaction pathway. Since the outputs of IA-MoE from different feature scales are further aggregated into the final representation F_u^{dec} for subsequent decoding and prediction, different feature levels may emphasize different interaction cues. Therefore, scale-specific routing preferences and a certain degree of routing concentration can be naturally expected, rather than indicating the degeneration of the overall adaptive interaction mechanism. Nevertheless, exploring how to further improve routing flexibility while preserving task-specific specialization remains an interesting direction. Future work will investigate more effective routing regularization and adaptive expert allocation strategies.

B. Failure Case Analysis and Limitations

Although SCD-MoE achieves a favorable tradeoff between performance and computational cost, some limitations still remain and warrant further investigation. First, the adaptive routing mechanism in MoE inevitably introduces additional routing computation and activation-memory overhead during training. Moreover, the current expert configuration, including the number of experts and the number of activated Top-K experts, is manually specified and fixed across datasets and feature levels. Such a fixed setting may not be optimal for scenes with different levels of semantic complexity. Future work will explore instance-aware and scene-adaptive mechanisms to dynamically adjust the expert capacity according to the complexity of the input.

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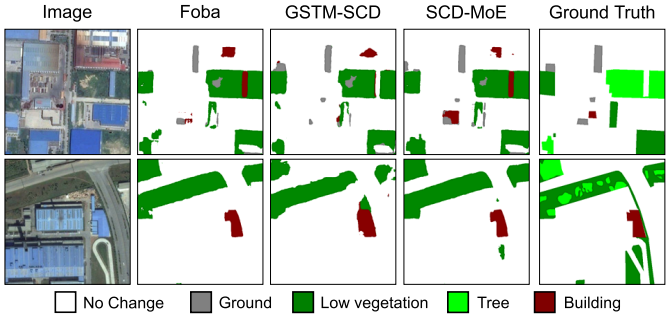


Fig. 11. Failure case analysis: semantic categories with similar visual appearances remain challenging for existing methods.

In addition, we examine representative failure cases to identify the limitations of our proposed method. As shown in Fig. 11, although the overall changed region is correctly identified, SCD-MoE and the competing methods confuse the tree and low vegetation categories. This example illustrates a broader challenge: when semantically distinct categories exhibit highly similar visual appearances, the available visual cues may be insufficient for reliable fine-grained semantic discrimination. Such semantic ambiguity represents a shared challenge for current SCD methods. Future work will investigate more discriminative semantic representations and ambiguity-aware learning objectives for visually similar land-cover categories.

VI. Conclusion

In this article, we proposed SCD-MoE, a compact dynamic framework for remote sensing SCD. The framework addresses two modeling limitations of existing methods. First, existing bi-temporal interaction methods do not adequately model sample-specific interaction preferences. Second, the distinct representation requirements of change localization and semantic recognition are often insufficiently differentiated. To address these issues, IA-MoE performs sample-adaptive routing over difference-aware and semantic-preserving experts, whereas TAR-MoE learns task-specific expert combinations from a shared expert pool, allowing task differentiation while retaining cross-task information sharing. Extensive experiments on SECOND, JL1, and Landsat-SCD show that SCD-MoE achieves SeK improvements ranging from 0.62 to 5.04 percentage points over the best-performing competing methods. Meanwhile, it requires only 48% of the parameters of the previous state-of-the-art method and substantially fewer FLOPs. Ablation studies, routing statistics, and feature analyses provide further evidence for the effectiveness of the proposed adaptive interaction and task-aware routing mechanisms. In future work, we plan to investigate adaptive expert allocation and more effective routing strategies to further explore the potential of MoE architectures for remote sensing change detection.

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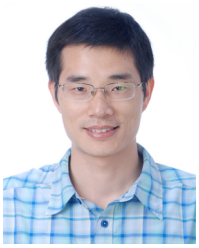
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